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Development of an Integrated System for Acoustic Calibration of Stethoscopes

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Degree in Electrical and Computer Engineering

Departamento de Engenharia Eletrotécnica
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October, 2026

This report partially satisfies the requirements defined for the Project/Internship course, in the 3rd year, of the Degree in Electrical and Computer Engineering.

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October, 2026

Abstract

Traditional acoustic stethoscopes frequently undergo silent material degradation, such as polyvinyl chloride (PVC) hardening and eartip fatigue, which severely attenuates clinically relevant cardiopulmonary sounds. Despite this diagnostic risk, these instruments remain largely exempt from routine metrological verification. This project aimed to develop a low-cost, integrated electromechanical calibration system capable of objectively characterizing stethoscope frequency response and signal fidelity. By utilizing a surface exciter coupled to a tissue-simulating elastomer phantom, the physical prototype successfully bypasses free-air acoustic decoupling to generate controlled, structure-borne vibrations across the 20 to 2000 Hz diagnostic bandwidth.

Data acquisition was driven by an ESP32 microcontroller sampling at 20,000 Hz, safely exceeding the Nyquist threshold, which fed a Python-based Short-Time Fourier Transform (STFT) spectral analysis pipeline. Controlled sinusoidal testing confirmed the successful capture of critical low-frequency cardiovascular targets at 20 Hz and 50 Hz, alongside high-frequency pulmonary signals up to 1111 Hz without attenuation or digital blindness. However, the spectral analysis also accurately identified system limitations, specifically nonlinear amplitude compression introduced by the pre-amplifier Automatic Gain Control (AGC) and Total Harmonic Distortion (THD) at 200 Hz caused by single-supply amplifier clipping.

Ultimately, the developed prototype successfully demonstrates a functional proof of concept for structure-borne stethoscope metrology. While analog hardware refinements are required to achieve perfect linearity, the system establishes a quantifiable, sustainable diagnostic tool designed to detect mechanical degradation and prevent compromised acoustic instruments from remaining in active clinical service.

Keywords: Stethoscope Calibration, Acoustic Transducer, Digital Signal Processing

Resumo

Os estetoscópios acústicos tradicionais sofrem frequentemente uma degradação material silenciosa, como o endurecimento do policloreto de vinilo (PVC) e a fadiga das olivas, o que atenua severamente os sons cardiopulmonares clinicamente relevantes. Apesar deste risco de diagnóstico, estes instrumentos permanecem, na sua maioria, isentos de verificação metrológica de rotina. Este projeto teve como objetivo desenvolver um sistema de calibração eletromecânico integrado e de baixo custo, capaz de caracterizar objetivamente a resposta em frequência e a fidelidade do sinal dos estetoscópios. Ao utilizar um excitador de superfície acoplado a um fantoma de elastómero que simula o tecido biológico, o protótipo físico contorna com sucesso o desacoplamento acústico em ar livre para gerar vibrações mecânicas estruturais controladas ao longo da largura de banda de diagnóstico de 20 a 2000 Hz.

A aquisição de dados foi controlada por um microcontrolador ESP32 a realizar amostragens a 20.000 Hz, superando com segurança o critério de Nyquist, o qual alimentou uma linha de processamento de análise espetral baseada na Transformada de Fourier de Curto Termo (STFT) em Python. Os testes sinusoidais controlados confirmaram a captação bem-sucedida de alvos cardiovasculares críticos de baixa frequência a 20 Hz e 50 Hz, juntamente com sinais pulmonares de alta frequência até 1111 Hz sem atenuação ou cegueira digital. No entanto, a análise espetral também identificou com precisão as limitações do sistema, especificamente a compressão de amplitude não linear introduzida pelo Controlo Automático de Ganho (AGC) do pré-amplificador e a Distorção Harmónica Total (THD) a 200 Hz causada pelo ceifamento (clipping) no amplificador de alimentação simples.

Em suma, o protótipo desenvolvido demonstra com sucesso uma prova de conceito funcional para a metrologia estrutural de estetoscópios. Embora sejam necessários refinamentos no hardware analógico para alcançar uma linearidade perfeita, o sistema estabelece uma ferramenta de diagnóstico quantificável e

sustentável, concebida para detetar a degradação mecânica e evitar que instrumentos acústicos comprometidos permaneçam no ativo em contexto clínico.

Palavras-Chave: Calibração de estetoscópios, transdutor acústico, processamento digital de sinais

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List of Symbols

Symbol	Description	Units
f	frequência	Hz
f	força	N
f_s	frequência de amostragem	Hz
i	corrente	A
M	massa	kg
P	potência	W
t	tempo	s
V	tensão elétrica	V
x	deslocamento	m
θ	deslocamento angular	rad
ω	velocidade angular	$\mathrm{rad\,s^{-1}}$

Chapter 1

Introduction

This report was developed within the scope of the Project/Internship course unit of the 3rd year of the Degree in Electrical and Computer Engineering (*Licenciatura em Engenharia Eletrotécnica e de Computadores* – LEEC) at the *Instituto Superior de Engenharia do Porto* (ISEP), during the 2025/2026 academic year.

The work focused on the design, development, and experimental validation of an integrated electroacoustic system for the objective calibration and frequency-response characterization of medical stethoscopes.

1.1 Problem Identification/Contextualization

Cardiovascular and pulmonary diseases remain among the leading causes of morbidity and mortality worldwide, requiring accurate, timely, and accessible diagnostic screening tools in clinical practice. The practice of clinical auscultation—listening to the internal biological sounds of the body—underwent a paradigm shift in 1816 with the invention of the first stethoscope by French

physician René Laennec [1]. Prior to this invention, physicians relied on immediate auscultation, placing their ear directly against the patient's chest, which posed significant acoustic and hygienic limitations. As illustrated in Figure 1.2, Laennec overcame this by channeling chest vibrations through a hollow cylinder, subsequently manufacturing the first formal monaural stethoscope from a turned wooden tube (Figure 1.1) [1].

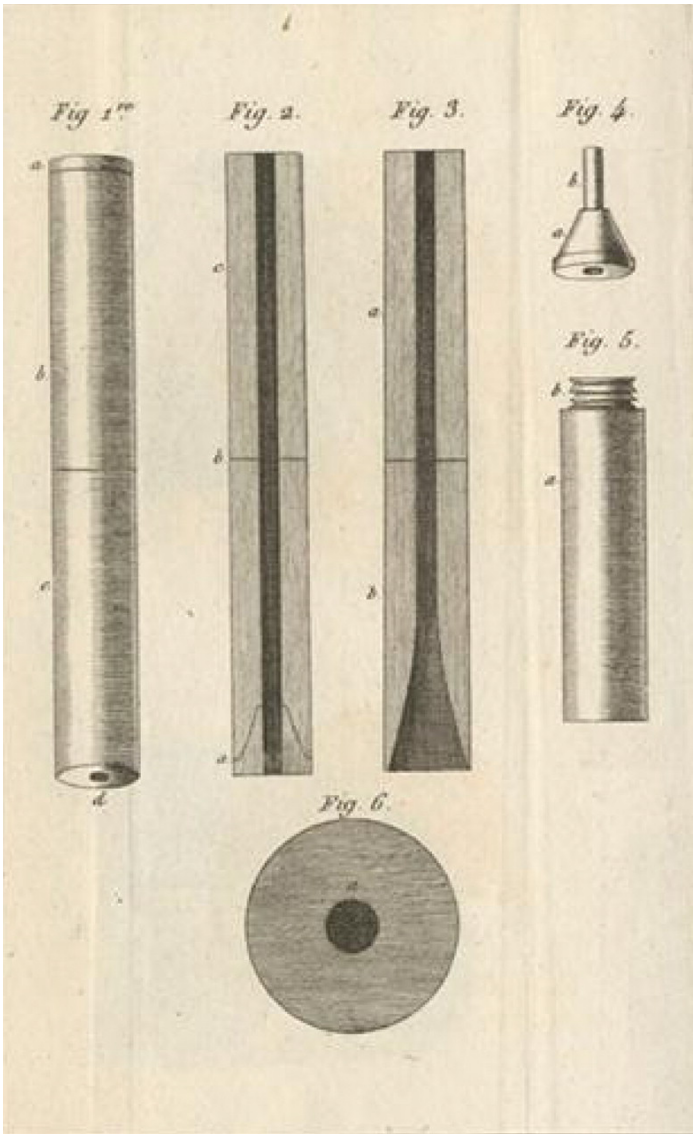


Figure 1.1: Original wooden monaural stethoscope designed by Laennec, showing its detachable plug and hollow acoustic lumen. Source: Adapted from Roguin [1].



Figure 1.2: Painting by Théobald Chartran depicting René Laennec performing auscultation at the Necker Hospital in 1816. Source: Adapted from Roguin [1].

Over the nineteenth and twentieth centuries, the instrument evolved through the introduction of flexible binaural tubing by Arthur Leared and George Cammann in 1851, and was later refined by Rappaport, Sprague, and David Littmann into the modern dual-head chestpiece architecture [2]. Today, traditional purely acoustic stethoscopes remain the dominant diagnostic instrument in global clinical practice due to their high mechanical durability, low acquisition cost, and complete independence from electrical power sources [3, 4].

However, despite two centuries of ergonomic refinement, the fundamental reliance on the passive mechanical transmission of acoustic pressure has remained unchanged. Consequently, purely acoustic auscultation is inherently constrained by two major factors: the subjective variability of human auditory perception and the unrecognized physical degradation of the instrument over time. Clinical studies have demonstrated that general practitioners miss up to 57% of significant valvular heart diseases when relying on standard acoustic stethoscopes [5]. While this high diagnostic error rate is frequently attributed to differences in clinical training or human auditory thresholds, a substantial portion of this signal loss stems from unnoticed material and mechanical failures within the stethoscope itself.

An acoustic stethoscope operates as a hermetically sealed, passive mechanical wave-guide that captures structure-borne vibrations from the patient's chest wall and channels them as acoustic pressure waves to the clinician's ear canals

[3]. The fidelity of this acoustic transmission depends entirely on the physical integrity of its constituent materials: the chestpiece diaphragm and bell, the internal acoustic lumen, the flexible polyvinyl chloride (PVC) tubing, and the elastomeric eartips.

In routine hospital and clinic environments, stethoscopes are subjected to harsh chemical and mechanical stressors. Strict nosocomial infection control protocols require clinicians to sanitize their instruments between every patient encounter using isopropyl alcohol wipes or chlorine-based disinfectants. Repeated exposure to these chemical solvents actively leaches plasticizers from the PVC polymer matrix, while continuous contact with human skin lipids (sebum), caused by draping the instrument around the neck, catalyzes polymer stiffening and surface micro-cracking [6, 7]. Simultaneously, diaphragms suffer from mechanical fatigue, and eartips lose their elastomeric compliance.

Because this material degradation occurs gradually over months and years of clinical service, it induces an insidious phenomenon: clinicians subconsciously adapt to the progressive loss of acoustic volume and spectral clarity, remaining entirely unaware that their instrument is compromised. When hardened tubing, micro-fissures, or compromised eartip seals alter the acoustic impedance of the pathway, critical low-frequency cardiovascular biomarkers, such as the pathological S3 and S4 ventricular gallops (20–50 Hz), which serve as primary indicators of congestive heart failure and ventricular hypertrophy, are severely attenuated before reaching the tympanic membrane [8]. Similarly, high-frequency pulmonary crackles and murmurs (200–2000 Hz) become masked by ambient room noise penetrating the rigid tubing walls [3].

This silent mechanical attenuation directly results in false-negative diagnoses, delayed medical interventions, and compromised patient safety. Furthermore, unlike electronic medical equipment, such as electrocardiographs, defibrillators, or infusion pumps, which undergo mandatory periodic metrological verification in hospital clinical engineering departments, acoustic stethoscopes are rarely, if ever, subjected to quantitative performance testing. Establishing an objective, repeatable, and low-cost electroacoustic calibration framework is therefore not merely a matter of instrument maintenance, but a vital clinical requirement to eliminate diagnostic blind spots and ensure patient safety.

1.1.1 PICO Framework

To structure the investigation and bridge the gap between clinical auscultation risks and electroacoustic engineering, the problem was formalized using the PICO (Population/Problem, Intervention, Comparison, Outcome) evidence-based framework, as summarized in Table 1.1.

Table 1.1: Formulation of the research scope using the PICO framework.

Problem / Population	Acoustic and electronic stethoscopes operating in clinical environments that are subject to gradual polymer degradation, diaphragm fatigue, and acoustic seal failures across the 20–2000 Hz cardiopulmonary bandwidth.
Intervention	Development of an integrated, low-cost electroacoustic calibration and testing rig combining structure-borne mechanical excitation, low-noise analog signal conditioning, high-speed microcontroller digitization, and Short-Time Fourier Transform (STFT) spectral analysis.
Comparison	Objective, quantitative spectral characterization against controlled reference sinusoidal sweeps (20–2000 Hz), contrasted with current subjective human auditory checks and uncalibrated clinical usage.
Outcome	Quantitative mapping of frequency response, verification of extreme low-frequency sensitivity (20–50 Hz), identification of Total Harmonic Distortion (THD) and noise-floor anomalies, and establishment of a repeatable metrological validation protocol.

1.2 Research Questions

Following the problem identification and the PICO formulation, a central research question was established to guide the engineering development of this project: *Is it possible to design and validate a low-cost, microcontroller-based electroacoustic testing rig capable of objectively characterizing the frequency response and signal fidelity of stethoscopes across the full* 20–2000 *Hz cardiopulmonary bandwidth?*

To systematically address this question, four specific research questions (RQ) were defined:

- **RQ1:** What are the primary physical and chemical degradation mechanisms in modern stethoscopes, and how do they acoustically alter the

transmission of critical cardiopulmonary biomarkers across low (20–50 Hz), intermediate (200 Hz), and high (1000–2000 Hz) frequency bands?

- **RQ2:** How can a low-cost Class-AB power amplifier and mechanical surface exciter be engineered to generate structure-borne acoustic signals that accurately replicate the frequency spectrum of human cardiopulmonary sounds, while maintaining linearity and minimizing Total Harmonic Distortion (THD) across the 20–2000 Hz diagnostic bandwidth?
- **RQ3:** To what extent do analog signal conditioning constraints introduce amplitude instabilities or Total Harmonic Distortion (THD) into the calibration baseline?

1.3 Objectives and Motivation

General Objective

To design, implement, and validate an integrated, electroacoustic calibration and spectral visualization system suitable for the objective characterization of stethoscope frequency response and signal fidelity across the 20–2000 Hz cardiopulmonary bandwidth.

Specific Objectives

Currently, the assessment of stethoscope integrity in clinical practice relies on subjective, ad-hoc auditory checks by healthcare professionals, allowing degraded instruments to remain in active rotation. To complement clinical practice with objective and verifiable engineering metrics, this work establishes the following specific objectives:

- **Investigate Stethoscope Degradation and Biomedical Acoustics:** Conduct a comprehensive literature review on stethoscope mechanical architecture, material degradation mechanisms, medical device metrology, and the acoustic signatures of physiological and pathological cardiopulmonary sounds.
- **Simulate Structure-Borne Cardiopulmonary Acoustics:** Study and implement an electroacoustic excitation stage and tissue-simulating

coupling medium capable of generating controlled vibrations that mimic the frequency spectrum and acoustic impedance of the human thorax.

- **Capture High-Fidelity Acoustic Data:** Evaluate and integrate an acoustic sensing, pre-amplification, and analog filtering topology capable of coupling to a stethoscope eartip and capturing faint low-frequency signals without introducing significant distortion or environmental noise.
- **Develop a High-Speed Digitization and Spectral Analysis Pipeline:** Identify and implement a digital acquisition and signal processing methodology capable of satisfying the Nyquist sampling criterion across the target medical bandwidth and computing time-frequency representations of the acquired audio.
- **Establish a Repeatable Validation Protocol:** Formulate a standardized testing protocol using controlled sinusoidal frequency sweeps to characterize system linearity, bandwidth limits, and harmonic distortion in a repeatable manner.

1.4 Project Schedule

Table 1.2 illustrates the Gantt chart distribution of the tasks performed.

Table 1.2: Project execution schedule and task distribution.

Task / Phase	W1	W2	W3	W4	W5	W6	W7	W8	W9	W10	W11	W12	W13	W14
1. Problem Definition & Scope														
2. Literature Review & State of the Art														
3. Architecture & Component Selection														
4. System Development & Testing														
5. Report Writing & Final Revision														

During the first two weeks (Weeks 1–2), the clinical problem was contextualized, defining the PICO parameters, research questions, and functional requirements for cardiopulmonary auscultation. Between Weeks 2 and 5, an extensive literature review was conducted covering stethoscope anatomy, polymer degradation chemistry, medical metrology, cardiopulmonary acoustic signatures, and transducer response characteristics.

Between Weeks 4 and 6, the system architecture was formalized by evaluating competing sound generation technologies, power amplification topologies, and acoustic sensing transducers. Subsequently, from Weeks 6 to 11, the system development and experimental validation were carried out, encompassing the assembly of the hardware prototype, the implementation of the digital acquisition and spectral processing software, and the execution of controlled sinusoidal frequency sweeps. In parallel with the final stages of experimental testing, Weeks 9 to 14 (following initial structuring in Week 1) were dedicated to data interpretation, the comprehensive drafting of the theoretical and technical chapters, and the final revision of the report.

1.5 Report Organization

This report is organized into six main chapters, structured as follows:

- **Chapter 1 – Introduction:** Contextualizes the clinical problem of stethoscope degradation, justifies the metrological importance of the work, defines the PICO framework, research questions, general and specific objectives, project schedule, and overall document structure.
- **Chapter 2 – Literature Review:** Presents the theoretical and scientific foundation of the project. It details the historical evolution and mechanical components of modern stethoscopes, physical and chemical sources of acoustic error, medical device metrology, the acoustic signatures of heart and lung sounds, frequency response curves of speakers and microphones, and existing academic and commercial validation systems.
- **Chapter 3 – Methodology:** Describes the engineering methodology, functional and non-functional requirements, electromechanical system architecture, hardware component selection, circuit schematics, mathematical filter calculations, microcontroller digitization parameters, and the Python DSP spectral analysis pipeline.
- **Chapter 4 – Results and Discussion:** Presents the experimental characterization of the prototype across controlled sinusoidal frequency sweeps (20 Hz, 50 Hz, 200 Hz, and 1111 Hz). It analyzes time-domain amplitude envelopes and STFT spectrograms, evaluates Automatic Gain Control (AGC) artifacts and Total Harmonic Distortion (THD), and

discusses the broader clinical and economic implications of stethoscope calibration versus electronic replacement.

- **Chapter 5 – Conclusion:** Synthesizes the main engineering contributions of the work, directly answers the proposed research questions, outlines the technical limitations encountered in the prototype, and establishes concrete directions for future development.
- **Chapter 6 – Appendices:** Contains the complete, commented source code listings developed for the ESP32 and the Python serial logging and spectral visualization algorithms.

Chapter 2

Literature Review

2.1 The Modern Stethoscope: Architecture and Error Sources

Despite the emergence of advanced medical imaging and digital point-of-care diagnostics, the acoustic stethoscope remains the primary screening instrument in daily clinical practice [3, 4]. To understand how physical wear and material aging compromise clinical auscultation, it is first necessary to examine the mechanical architecture of the modern stethoscope and the acoustic function of each of its constituent components.

2.1.1 Mechanical Architecture and Component Functions

A modern binaural acoustic stethoscope operates as a passive, hermetically sealed mechanical-to-acoustic impedance transformer and waveguide [8, 3]. Rather than generating energy, the instrument captures kinetic vibrations from the surface of the patient's skin, converts them into oscillating air pressure waves within a confined column of air, and channels those pressure waves

directly to the clinician's tympanic membranes. As established through the twentieth-century refinements of Rappaport, Sprague, and Littmann [2], the standard clinical stethoscope is divided into five primary functional assemblies:

1. **Chest Piece:** The dual-sided metallic head (typically machined from stainless steel, titanium, or anodized aluminum) that contacts the patient's thorax to capture structure-borne body sounds [2, 9]. It houses an internal rotating stem mechanism that allows the clinician to switch between the concave bell collector and the flat diaphragm collector.

2. **Bell Side:** The smaller, concave cup-shaped side of the chestpiece designed to capture low-frequency cardiovascular sounds (20–150 Hz), such as diastolic rumbles and the pathological S3 and S4 ventricular gallops [8]. It is fitted with a soft elastomeric non-chill rim for patient comfort, minimizing skin temperature sensation while forming an acoustic seal [9]. When applied with light mechanical pressure, the circumscribed area of human skin acts as a compliant, low-tension membrane; conversely, excessive downward force stretches the skin taut, raising its resonant frequency and attenuating low-frequency signals [8].

3. **Diaphragm Side:** The wider, flat side of the chestpiece engineered to isolate higher-frequency cardiopulmonary acoustics (100–2000 Hz), such as S1/S2 valvular splitting, systolic murmurs, and vesicular or adventitious lung sounds [2, 3, 9].

4. **Diaphragm Membrane:** The thin, flat, semi-rigid circular disc (typically manufactured from epoxy-fiberglass composites, polycarbonate, or polyurethane) seated across the diaphragm side [9]. Because the stiff membrane possesses a higher natural mechanical resonance, it acts as a mechanical high-pass filter—attenuating high-amplitude, low-frequency ventricular vibrations that would otherwise mask faint high-pitched pulmonary or vascular sounds [8, 3].

5. **Diaphragm Retaining Ring:** The precision-fitted elastomeric or metallic rim that locks the diaphragm membrane securely around the circumference of the chestpiece under uniform radial tension, preventing air leakage and non-linear edge rattling [9].

6. **PVC Tube:** The flexible acoustic transmission tube extruded from plasticized medical-grade polyvinyl chloride (PVC) [6, 9]. Its internal lumen

(typically 3–5 mm in diameter and 55–70 cm in length) channels pressure oscillations from the chestpiece to the headset, while its thick outer walls provide acoustic mass to reject ambient clinical noise [8].

7. **Binaural Spring:** The internal tension leaf-spring mechanism integrated within the upper yoke of the tubing [9]. It applies calibrated inward clamping force to keep the headset securely seated during auscultation while allowing ergonomic adjustment across different head sizes.

8. **Binaural Tube:** The two rigid, angled metallic tubes (typically aluminum or stainless steel) extending from the upper yoke that bifurcate the acoustic wave and direct sound into both ears [9]. They are anatomically canted forward at approximately 15° to align with the human external auditory canal.

9. **Ear Tip:** The soft, compliant silicone or synthetic rubber tips mounted at the ends of the binaural tubes [9]. They conform to the geometry of the clinician's ear canals to establish an airtight, hermetic acoustic seal—a critical requirement for preserving low-frequency pressure waves down to 20 Hz and blocking external room noise [8, 3].

Figure 2.1 illustrates the complete mechanical anatomy of a modern binaural acoustic stethoscope, detailing the chestpiece profile, diaphragm assembly, and binaural transmission tubing.

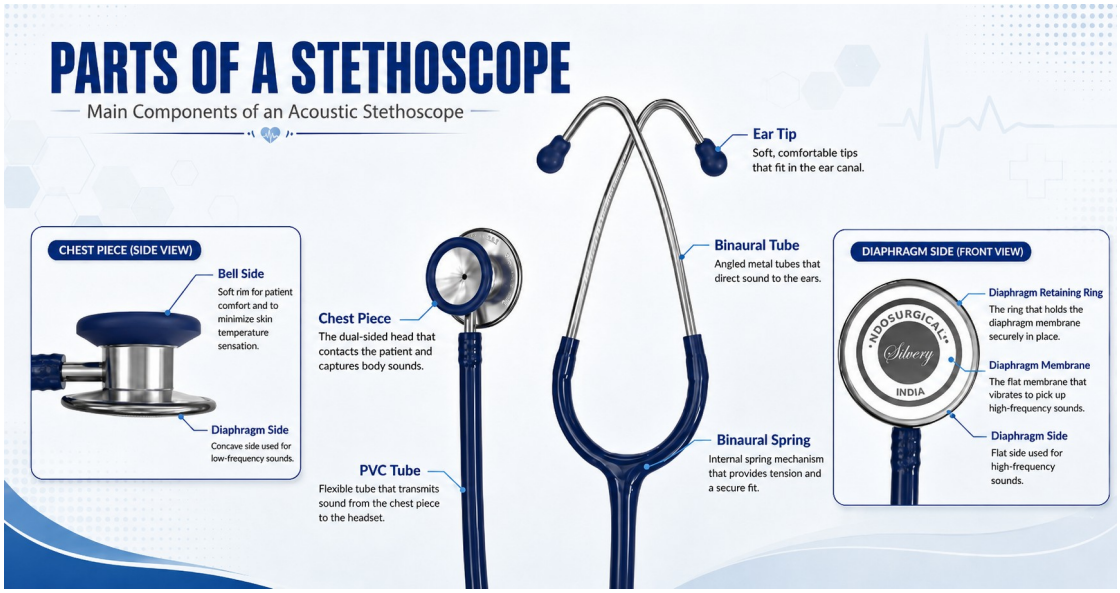


Figure 2.1: Main mechanical components of a dual-head acoustic stethoscope, detailing the side profile of the bell and diaphragm chestpiece (left) and the front view of the diaphragm retaining assembly (right). Source: Adapted from IndoSurgicals [9].

Table 2.1 synthesizes the mechanical components of a modern acoustic stethoscope, their standard materials of construction, and their respective roles in the acoustic transmission pathway.

Table 2.1: Summary of modern acoustic stethoscope components and their functional roles.

Component	Typical Material	Primary Acoustic Function	Target Band
Chest Piece	Stainless Steel, Titanium, or Aluminum	Dual-sided acoustic collector that contacts the thorax and routes vibrations via an internal stem valve.	20–2000 Hz
Bell Side	Machined Metal with Elastomer Rim	Concave low-impedance cup using compliant skin as a membrane to capture low-pitched heart sounds.	20–150 Hz
Diaphragm Side	Machined Metal Collector Face	Flat collector face housing the stiff membrane to isolate high-frequency cardiopulmonary sounds.	100–2000 Hz
Diaphragm Membrane	Epoxy-Fiberglass or Polycarbonate	Semi-rigid vibrating disc acting as a mechanical high-pass filter to prevent low-frequency masking.	100–2000 Hz
Diaphragm Retaining Ring	Molded Elastomer or Neoprene	Secures the diaphragm membrane under uniform radial tension and seals the perimeter edge.	100–2000 Hz
PVC Tube	Plasticized Medical PVC	Flexible, thick-walled waveguide that transmits acoustic pressure waves while rejecting ambient noise.	20–2000 Hz
Binaural Spring	Tempered Spring Steel	Internal tension mechanism that provides clamping force for a secure, airtight eartip fit.	Mechanical Support
Binaural Tube	Aluminum or Stainless Steel Alloy	Rigid angled tubes (15°) that split the acoustic path and direct sound waves toward both ear canals.	20–2000 Hz
Ear Tip	Soft Silicone or Synthetic Rubber	Forms a hermetic seal within the ear canal to prevent low-frequency pressure leakage and room noise ingress.	20–2000 Hz

2.1.2 Chemical and Mechanical Degradation in Clinical Environments

While traditional acoustic stethoscopes are engineered for mechanical durability, extensive polymer science literature demonstrates their vulnerability to progressive chemical and physical degradation over time. In its pure, uncompounded state, polyvinyl chloride (PVC) is a rigid, brittle glassy polymer with a glass transition temperature (T_g) near 80 °C. To achieve the flexibility, acoustic damping, and mechanical compliance required for stethoscope tubing, manufacturers compound medical-grade PVC with 30% to 40% by weight of low-molecular-weight plasticizers, such as di(2-ethylhexyl) phthalate (DEHP) or non-phthalate esters [6]. Because these plasticizer molecules are physically interspersed between the PVC polymer chains via weak secondary van der Waals forces rather than permanent covalent bonds, they are highly susceptible to phase migration and leaching when exposed to external chemical stressors.

In daily clinical practice, two primary mechanisms accelerate the breakdown of the PVC polymer matrix:

- **Solvent-Induced Plasticizer Leaching:** To prevent healthcare-associated infections (HAIs), clinical guidelines mandate disinfecting stethoscopes between patients using 70% isopropyl alcohol (IPA) pads, ethanol solutions, or quaternary ammonium compounds. Research into medical-grade PVC polymers by Josh et al. [6] demonstrates that frequent exposure to organic solvents aggressively accelerates the diffusion and extraction of essential plasticizers out of the polymer matrix. As plasticizer concentration drops, the polymer chains pack more tightly together, causing a progressive increase in the material's Young's modulus (stiffening), volume shrinkage, and eventual embrittlement.
- **Sebum and Lipophilic Fluid Absorption:** Clinicians routinely carry stethoscopes draped directly around the skin of the neck, exposing the upper yoke of the PVC tubing to continuous body heat (37 °C) and human skin lipids. Studies by Zhang et al. [7] establish that human sebum is a complex lipophilic mixture composed of triglycerides, free fatty acids, wax esters, and squalene. Because plasticizers are similarly lipophilic, sebum acts as a chemical extraction medium at the polymer

surface while simultaneously penetrating the micro-voids of the PVC matrix [7]. Over extended clinical service, this combined thermal and lipid exposure induces polymer hardening, loss of viscoelasticity, and the propagation of surface micro-cracking.

While the chemical degradation of medical PVC is thoroughly documented in polymer science, its direct quantitative correlation with acoustic transmission loss in diagnostic instruments remains largely underexplored in clinical maintenance protocols.

2.1.3 Sources of Acoustic Error and Transmission Loss

The structural integrity of each mechanical component is the primary variable governing the acoustic pressure transfer function of a stethoscope. Foundational acoustic studies by Ertel et al. [8] established that stethoscope transmission is not perfectly flat, but rather exhibits complex resonance peaks and attenuation valleys that are highly sensitive to structural defects. Furthermore, Leng et al. [3] highlighted that physical degradation along the acoustic pathway drastically reduces the Signal-to-Noise Ratio (SNR), effectively masking faint physiological frequencies beneath ambient hospital noise.

From an acoustic metrology perspective, instrument degradation introduces three distinct sources of diagnostic error across the cardiopulmonary spectrum:

1. **Loss of Hermeticity and Low-Frequency Attenuation (20–100 Hz):** The transmission of extreme low-frequency cardiovascular sounds, such as the S1 and S2 fundamentals and pathological S3/S4 gallops, relies on piston-like, closed-chamber pressure variations within the stethoscope lumen [8]. If the elastomeric eartips harden and fail to seal against the ear canal, if the rotating stem valve O-ring wears down, or if a micro-fissure penetrates the hardened PVC tubing wall, the hermetic seal is broken. Acoustically, an air leak acts as a shunt impedance to the surrounding atmosphere, venting slow pressure oscillations before they can displace the tympanic membrane. Even a pinhole leak or a poorly seated eartip can induce severe attenuation exceeding 15 to 20 dB below 100 Hz, rendering low-frequency heart failure biomarkers completely inaudible [8, 3].

2. **Tubing Wall Stiffening and Ambient Noise Penetration:** In an intact stethoscope, the compliant, viscoelastic PVC tubing walls absorb

external airborne noise and dampen internal wall vibrations. As plasticizer depletion increases the stiffness of the tubing [6], its mechanical impedance changes. Hardened tubing acts like a conductor, transmitting friction noise from handling (such as fingers or clothing rubbing against the tube) directly into the lumen, but also loses its acoustic isolation properties, allowing ambient ward noise (50–1000 Hz) to penetrate the waveguide and degrade the SNR [3].

3. **Diaphragm Fatigue and Resonance Shifting (**200–2000 **Hz):** The chestpiece diaphragm is subjected to repeated mechanical flexing, accidental impacts, and chemical wiping. Micro-fractures, warping of the epoxy-fiberglass disc, or loosening of the retaining chill-ring alters the mechanical tension of the membrane. This shifts the diaphragm's natural resonant frequency and introduces non-linear mechanical rattling or harmonic distortion, compromising the faithful capture of high-frequency pulmonary crackles, wheezes, and high-pitched valvular murmurs [3].

Despite these well-defined mechanical failure modes and their severe diagnostic risks, healthcare institutions lack standardized, objective calibration rigs capable of quantifying these acoustic errors.

2.2 Medical Metrology and Biomedical Acoustics

The reliable quantification and performance verification of diagnostic medical devices rely upon the established principles of medical metrology and biomedical acoustics. While general engineering metrology governs standardized measurement units and instrument traceability, medical metrology extends these requirements to clinical environments to ensure that diagnostic equipment maintains measurement accuracy, repeatability, and patient safety throughout its operational lifecycle [10, 11].

2.2.1 Metrology and Quality Assurance in Medical Devices

In modern healthcare infrastructure, medical instruments that directly influence clinical decision-making are subject to rigorous quality assurance (QA)

frameworks and periodic metrological auditing [10]. Electronic diagnostic and therapeutic equipment, such as electrocardiographs, defibrillators, infusion pumps, and audiometers, must undergo routine verification against traceable reference standards to detect calibration drift, electronic component aging, or transducer degradation [10, 11].

However, a significant metrological gap persists regarding clinical auscultation. While active electronic monitors are governed by strict preventive maintenance schedules, traditional acoustic stethoscopes are classified as passive mechanical devices and are consequently exempted from routine quantitative verification. Because unrecognized material degradation in a stethoscope introduces severe acoustic attenuation that contributes to missed diagnoses of valvular and cardiopulmonary pathologies [8, 5], establishing an objective metrological testing protocol is essential for clinical quality assurance. A valid electroacoustic calibration system must provide a controlled, repeatable excitation source and an accurate acoustic reference sensor to quantify deviations from baseline transmission characteristics [12].

2.2.2 Principles of Acoustic Impedance and Boundary Reflection

Biomedical acoustics describes the generation, propagation, and attenuation of mechanical pressure waves through biological tissues and diagnostic waveguides [11, 13]. As an acoustic wave travels through any physical medium, its propagation behavior is governed by the medium's characteristic acoustic impedance, which represents the material's inherent resistance to acoustic particle motion based on its density and elasticity [14, 11].

When a sound wave traveling through one medium encounters a boundary with a second medium of a different acoustic impedance, only a fraction of the acoustic energy is transmitted across the interface, while the remaining energy is reflected back toward the source [14]. The greater the difference in acoustic impedance between the two materials, the greater the percentage of sound energy that is lost to reflection at the boundary [11].

In clinical auscultation, cardiovascular and pulmonary sounds originate deep

within the thorax, which is composed predominantly of water-rich soft tissues, myocardium, and blood. Because biological tissue is dense and relatively incompressible, it possesses a very high characteristic acoustic impedance [11, 13]. In contrast, ambient air at room temperature has a very low density and high compressibility, resulting in an acoustic impedance that is orders of magnitude lower than that of human tissue [14, 11].

Because of this extreme impedance mismatch between the dense thoracic wall and the surrounding air, over 99% of the internal acoustic energy generated by the heart and lungs is reflected back into the body when it reaches the skin surface, with less than 1% radiating into the open air [11, 13].

2.3 Acoustic Signatures of Cardiopulmonary and Biological Sounds

To evaluate the acoustic performance of a stethoscope objectively, it is essential to define the frequency spectrum and amplitude characteristics of the biological signals the instrument is designed to capture. The human body generates a wide variety of mechanical vibrations resulting from myocardial contraction, valvular closure, turbulent blood flow, airflow through the respiratory tract, and gastrointestinal motility [3, 11]. Together, these acoustic signatures span a total diagnostic bandwidth from 20 Hz to 2000 Hz [8, 3].

2.3.1 Cardiovascular Acoustic Signatures

Heart sounds are transient, low-frequency mechanical vibrations generated by the acceleration and deceleration of blood and the tensioning of cardiac valves during the cardiac cycle [3, 11]. Because the majority of cardiac acoustic energy is concentrated below 200 Hz, where the human ear exhibits naturally reduced sensitivity, preserving low-frequency transmission is critical for cardiovascular diagnosis [8, 3].

First and Second Heart Sounds (S1 and S2) In a healthy heart, the standard auscultatory rhythm consists of two primary physiological sounds, commonly described as the “lub-dub” cycle [3]:

- **First Heart Sound (S1):** Occurs at the onset of ventricular systole and is primarily produced by the closure of the mitral and tricuspid

(atrioventricular) valves as ventricular pressure rises [11]. Acoustically, S1 is a dull, low-pitched sound lasting approximately 100 to 150 ms, with its primary spectral energy concentrated between 20 Hz and 150 Hz [3].

- **Second Heart Sound (S2):** Marks the end of ventricular systole and the beginning of diastole, resulting from the closure of the aortic and pulmonic (semilunar) valves [11]. Because the semilunar valves close under higher arterial pressure and possess stiffer leaflets, S2 is shorter in duration and slightly higher in pitch than S1, with dominant frequencies spanning 50 Hz to 250 Hz [3].

Third and Fourth Heart Sounds (S3 and S4 Gallops) During ventricular diastole, two additional low-frequency sounds may occur, creating a cadence that resembles the gallop of a horse [3]:

- **Third Heart Sound (S3):** Occurs in early diastole during rapid passive filling of the ventricles. While occasionally physiological in children or young athletes, an S3 gallop in older adults is a primary clinical indicator of ventricular dilation and congestive heart failure [3, 11].
- **Fourth Heart Sound (S4):** Occurs in late diastole when the atria contract forcefully against a stiff, non-compliant ventricle, serving as a marker of ventricular hypertrophy or myocardial ischemia [3].

Both S3 and S4 are extremely faint, low-energy vibrations located at the very bottom of the human auditory spectrum, between 20 Hz and 50 Hz (occasionally extending up to 70 Hz) [8, 3]. Because their amplitude sits near the human hearing threshold, any loss of hermeticity in a stethoscope's eartips or hardening of its PVC tubing renders S3 and S4 gallops completely inaudible [8].

Cardiac Murmurs Unlike the brief, transient clicks of valvular closure, cardiac murmurs are longer, noisy acoustic events caused by turbulent blood flow across narrowed (stenotic) or leaking (regurgitant) heart valves, or through septal defects [3, 11]. Because fluid turbulence generates higher-frequency oscillations, murmurs span a much broader spectrum, typically ranging from 100 Hz to 600 Hz, with high-velocity regurgitant murmurs occasionally reaching 1000 Hz [3].

2.3.2 Pulmonary Acoustic Signatures

Respiratory sounds are generated by turbulent and vortex airflow moving through the tracheobronchial tree and alveoli during inspiration and expiration [13]. Because the lungs and chest wall act as a natural low-pass filter that attenuates high frequencies as sound travels toward the skin, pulmonary auscultation requires a stethoscope capable of capturing higher frequencies up to 2000 Hz [13, 3].

Normal Breath Sounds (Vesicular and Tracheal) Normal respiratory sounds are classified according to the anatomical region where they are auscultated [13]:

- **Vesicular Breath Sounds:** Heard over the majority of the peripheral lung fields during inspiration and early expiration. Because the sound waves must travel through air-filled alveoli and soft tissue, higher frequencies are dampened, resulting in a soft, low-pitched rustling sound with dominant energy between 100 Hz and 500 Hz [13, 3].
- **Bronchial and Tracheal Sounds:** Heard directly over the sternum, large central airways, and trachea. With minimal tissue filtering between the airway and the stethoscope diaphragm, these sounds are louder, harsher, and contain a wide frequency spectrum extending from 100 Hz up to 1500 Hz [13].

Adventitious Lung Sounds (Wheezes, Rhonchi, and Crackles) When pulmonary pathologies alter airway geometry or introduce fluid into the lungs, abnormal (adventitious) sounds are superimposed over normal breath sounds [13, 3]:

- **Wheezes and Rhonchi:** Continuous, musical tones caused by airflow passing through narrowed or partially obstructed airways, which induces flutter in the bronchial walls (commonly seen in asthma, bronchitis, or chronic obstructive pulmonary disease) [13]. Low-pitched snoring sounds (rhonchi) concentrate between 100 Hz and 300 Hz, whereas high-pitched whistling wheezes span from 400 Hz up to 1600 Hz [13, 3].
- **Crackles (Rales):** Discontinuous, brief, explosive clicking or bubbling sounds lasting less than 20 ms [13]. They are generated by the sudden

snapping open of collapsed small airways during inspiration or by air bubbling through pulmonary secretions in conditions such as pneumonia, pulmonary edema, and interstitial fibrosis [13]. Coarse crackles occupy lower frequencies around 100 to 500 Hz, while fine crackles contain high-frequency components reaching up to 2000 Hz [13, 3].

2.3.3 Other Clinical Auscultation Sounds

Beyond cardiopulmonary assessment, clinical stethoscopes are routinely utilized to evaluate vascular and gastrointestinal acoustics [3, 11]:

- **Korotkoff Sounds and Vascular Bruits:** During manual blood pressure measurement with a sphygmomanometer cuff, the clinician listens over the brachial artery for Korotkoff sounds, which are low-frequency tapping and turbulent flow sounds (20–300 Hz) triggered by pulsatile blood flow through the compressed artery [11]. Similarly, auscultation over the carotid, renal, or femoral arteries is performed to detect vascular bruits (100–500 Hz), which indicate atherosclerotic plaque narrowing [3].
- **Gastrointestinal (Bowel) Sounds:** Abdominal auscultation captures the movement of gas and fluid propelled by peristalsis. Normal and hyperactive bowel sounds consist of irregular clicks and gurgles spanning 100 Hz to 1000 Hz, whose absence can indicate paralytic ileus or intestinal obstruction [3].

Table 2.2 summarizes the primary physiological and pathological biological sounds, their clinical mechanisms, and their respective frequency bandwidths, establishing why a comprehensive stethoscope calibration system must evaluate the full 20 Hz to 2000 Hz spectrum.

2.3.4 Electroacoustic Transducer Frequency Response Curves

To generate and capture the cardiopulmonary acoustic signatures summarized in Table 2.2, an objective calibration system relies on electroacoustic transducers: an output actuator (loudspeaker or mechanical exciter) to synthesize the biological vibrations, and an input sensor (microphone) to measure the transmitted wave at the stethoscope eartips [3, 12]. The accuracy of this measurement is governed by the frequency response curve of each transducer, which

Table 2.2: Summary of cardiopulmonary and biological acoustic signatures and their clinical frequency ranges.

Biological Sound	Anatomical System	Frequency Range	Clinical Mechanism and Significance
S3 and S4 Gallops	Cardiovascular	20–70 Hz	Faint diastolic ventricular filling vibrations; key markers of heart failure and stiff ventricles [8, 3].
First Heart Sound (S1)	Cardiovascular	20–150 Hz	Closure of mitral and tricuspid valves at the start of ventricular systole [3, 11].
Second Heart Sound (S2)	Cardiovascular	50–250 Hz	Closure of aortic and pulmonic valves marking the onset of ventricular diastole [3, 11].
Korotkoff Sounds	Vascular	20–300 Hz	Pulsatile arterial tapping during blood pressure cuff deflation [11].
Cardiac Murmurs	Cardiovascular	100–1000 Hz	Turbulent blood flow across stenotic or regurgitant heart valves [3].
Vesicular Breath Sounds	Pulmonary	100–500 Hz	Normal peripheral alveolar airflow filtered by the thoracic wall [13, 3].
Tracheal Breath Sounds	Pulmonary	100–1500 Hz	Unfiltered turbulent airflow heard over the trachea and central airways [13].
Rhonchi and Wheezes	Pulmonary	100–1600 Hz	Continuous musical oscillations from narrowed airways in asthma or COPD [13, 3].
Pulmonary Crackles	Pulmonary	100–2000 Hz	Explosive reopening of collapsed airways or fluid bubbling in pneumonia and edema [13, 3].

plots its output sensitivity (in decibels, dB) across the operating frequency spectrum [15, 16].

An ideal metrological transducer exhibits a completely flat frequency response curve (0 dB deviation) across the entire 20 Hz to 2000 Hz medical bandwidth, ensuring that every physiological frequency is reproduced or measured with equal weight [12, 15]. In practice, however, physical resonance limits and mechanical damping cause real-world speakers and microphones to deviate significantly from a flat response, particularly at the low-frequency boundary (20 to 100 Hz) where S1, S2, and S3/S4 cardiac sounds reside [3, 16].

Sound Generation Curves: Dynamic Speakers vs. Surface Exciters Conventional dynamic loudspeakers generate airborne sound by driving a suspended paper or polymer cone with an electromagnetic voice coil [16]. The frequency response curve of a dynamic speaker is divided into three distinct regions bounded by its fundamental mechanical resonant frequency (f_0) [16]:

- **High-Frequency Roll-Off Zone:** Above the working band, the mass and inductance of the voice coil limit rapid acceleration, causing acoustic output to drop at higher frequencies.
- **Linear Piston Band:** Above the fundamental resonance, the speaker cone operates in a relatively flat, linear region where output sound pressure remains proportional to the electrical input signal.
- **Low-Frequency Roll-Off Zone (Below f_0):** Below the speaker's natural resonant frequency, the mechanical stiffness of the suspension spider and the small surface area of the cone prevent effective air displacement, causing the frequency response curve to drop steeply at a rate of 12 dB per octave [16].

In compact, portable loudspeakers (25 to 50 mm diameter), the fundamental resonance typically sits between 150 Hz and 300 Hz [16]. Consequently, their free-air frequency response curve suffers severe attenuation across the 20 to 100 Hz cardiac band. In contrast, a frameless mechanical surface exciter eliminates the moving air cone entirely and couples its voice coil directly to a solid substrate or viscoelastic phantom [12]. By driving mechanical vibrations directly into the damped phantom material rather than displacing open air,

a surface exciter maintains usable low-frequency energy down to 20 Hz while extending smoothly past the 2000 Hz pulmonary ceiling [12].

Figure 2.2 presents the measured frequency response curve of the Dayton Audio DAEX25FHE-4 structural exciter transposing acoustic energy through a rigid substrate, plotted in decibels of Sound Pressure Level (dB SPL) relative to the 20 to 2000 Hz cardiopulmonary auscultation spectrum.

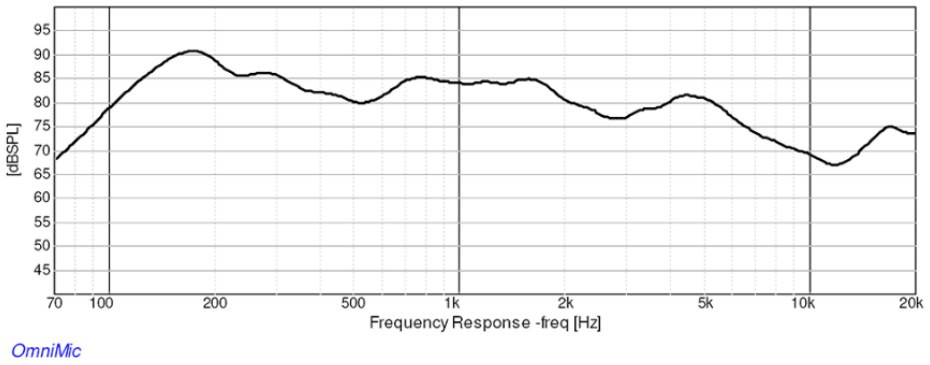


Figure 2.2: Measured frequency response of the Dayton Audio DAEX25FHE-4 structural exciter in dB SPL, featuring the primary mechanical resonance peak within the lower spectrum boundary. Source: Adapted from Dayton Audio [17].

Sound Acquisition Curves: Electret Condenser (ECM) vs. MEMS Microphones At the receiving end of the stethoscope, the acoustic pressure wave exiting the eartip must be converted into an electrical voltage by a microphone capsule [3]. In modern instrumentation, two primary microphone architectures are utilized, each exhibiting a distinct low-frequency response curve [15]:

- **Omnidirectional Electret Condenser Microphones (ECM):** An ECM consists of a flexible metallized polymer diaphragm positioned parallel to a permanently charged backplate [15]. Because the capsule responds directly to absolute acoustic pressure variations, particularly when coupled into a sealed acoustic cavity such as a stethoscope eartip, an omnidirectional ECM exhibits an exceptionally flat frequency response curve extending from 20 Hz up to $15,000$ Hz (within ± 2 dB), making it the gold-standard sensor for phonocardiography [3, 12, 15].

- **Micro-Electro-Mechanical Systems (MEMS) Microphones:** MEMS sensors are miniature silicon-etched capacitive transducers widely designed for consumer voice communications in smartphones and wearable electronics. Because human speech contains minimal energy below 100 Hz, standard consumer MEMS capsules incorporate microscopic pressure-equalization vents in the silicon membrane to prevent barometric overload and wind noise. Acoustically, this vent acts as a mechanical high-pass filter, introducing a steep low-frequency roll-off below 60 to 100 Hz that severely attenuates the 20 to 50 Hz S3 and S4 cardiac gallop band [3, 15].

Figure 2.3 illustrates the characteristic normalized frequency response curves of an omnidirectional ECM capsule, a standard consumer MEMS microphone, and a compact free-air dynamic speaker relative to the 20 to 2000 Hz cardiopulmonary auscultation spectrum.

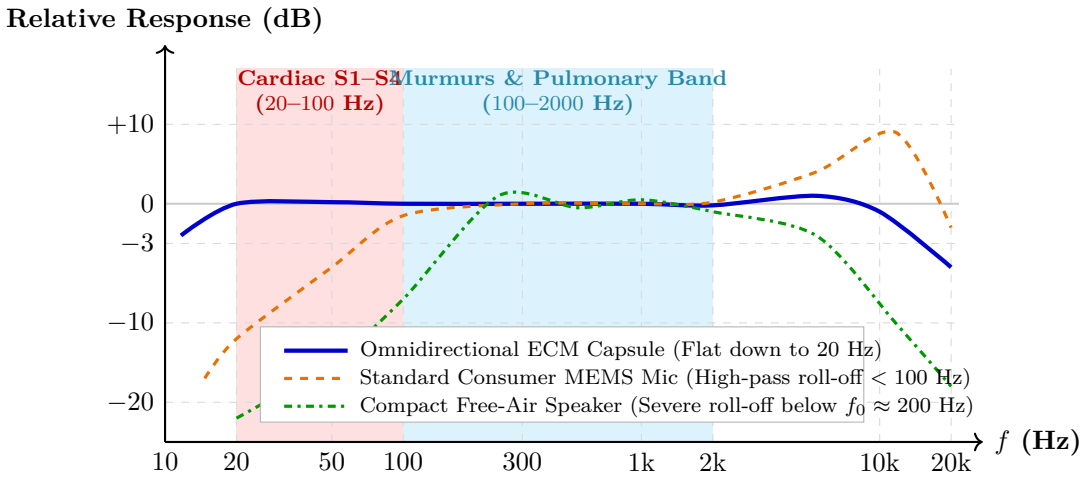


Figure 2.3: Comparison of typical normalized frequency response curves for an omnidirectional ECM capsule, a standard consumer MEMS microphone, and a compact free-air loudspeaker across the cardiopulmonary auscultation bandwidth. Source: Adapted from Eargle [15], Ballou [16], and Lewis [18].

2.4 Existing Stethoscope Validation Systems and Approaches

Despite the recognized clinical risks associated with stethoscope degradation, the development of standardized, clinically accessible validation systems remains a largely unresolved challenge in biomedical engineering. The current state of the art in stethoscope metrology can be divided into high-fidelity industrial characterization methods and emerging academic prototypes, both of which present distinct limitations for routine hospital deployment.

2.4.1 Industrial and Laboratory Reference Standards

At the manufacturer level, the acoustic frequency response of a new stethoscope design is typically evaluated using laboratory-grade audiometric testing equipment. These setups frequently involve an artificial chest phantom constructed from viscoelastic polymers, coupled to a calibrated mechanical vibration exciter [13]. The stethoscope under test is mounted to the phantom under a precisely controlled static clamping force, and the output at the eartips is measured using laboratory reference microphones [12].

While these industrial systems provide exceptional metrological accuracy and repeatability, they are prohibitively expensive, physically bulky, and strictly confined to specialized acoustic laboratories. Consequently, they are utilized exclusively for initial product research and development rather than the routine periodic maintenance of active clinical inventory.

2.4.2 Academic Validation Prototypes and Methodologies

In recent years, biomedical research groups have proposed custom electroacoustic test rigs designed to validate the frequency response of stethoscopes outside of industrial environments. The most comprehensive modern methodology was detailed by Klum et al. (2019), who developed a dedicated validation system specifically for evaluating digital stethoscopes [12].

To overcome the severe acoustic impedance mismatch of free-air coupling, Klum et al. utilized an electro-dynamic surface exciter bonded to a proprietary silicone elastomer matrix, which successfully simulated the acoustic

impedance and wave propagation characteristics of human soft tissue. By driving the exciter with logarithmic sine sweeps and recording the acoustic output with an omnidirectional electret condenser microphone (ECM) sealed to the stethoscope eartip, the system successfully quantified the low-frequency transfer functions of various digital models down to the critical 20 Hz limit [12].

Other academic approaches have focused purely on electronic signal injection. In these studies, researchers bypass the physical stethoscope chestpiece entirely by injecting synthetic cardiopulmonary audio files directly into the analog-to-digital converter of an electronic stethoscope [3]. While this approach allows for rigorous testing of internal digital filtering and amplification algorithms, it completely ignores the passive mechanical acoustic pathway. Because the majority of real-world instrument failures stem from hardened PVC tubing, cracked diaphragms, or degraded eartip seals [8, 3], purely electronic validation is incapable of identifying structurally compromised diagnostic tools.

2.4.3 Identification of the Technological Gap

A review of the current literature reveals a clear technological gap in medical device metrology. Industrial HATS calibration rigs are too costly and complex for hospital engineering departments, while purely electronic signal injection methods fail to assess mechanical hardware degradation. Furthermore, highly accurate structure-borne testing rigs, such as the system developed by Klum et al., have historically focused on digital stethoscopes and require separate benchtop signal generators, external audio interfaces, and external computers for data analysis [12].

Chapter 3

Methodology

3.1 System Requirements

To design an integrated calibration rig capable of characterizing acoustic and electronic stethoscopes, the system must satisfy a rigorous set of functional requirements (defining operational capabilities) and non-functional requirements (defining performance constraints and signal fidelity).

3.1.1 Functional Requirements

The functional requirements outline the necessary signal generation, mechanical coupling, and digital processing capabilities of the calibration rig, as detailed in Table 3.1.

Table 3.1: Functional requirements for the integrated system.

Requirement	**Engineering Description**
Controlled Acoustic Excitation	The system must generate continuous logarithmic sine sweeps covering the full cardiopulmonary frequency spectrum (20 Hz to 2000 Hz).
Structure-Borne Vibration	Acoustic energy must be delivered via a surface exciter mechanically coupled to an elastomeric tissue-simulating medium, avoiding free-air acoustic impedance mismatch.
Acoustic Sealing & Acquisition	The acquisition sensor must form an airtight seal with the stethoscope eartip to prevent low-frequency attenuation and accurately capture cardiopulmonary acoustics down to 20 Hz.
Signal Pre-Amplification	The low-amplitude analog output from the microphone capsule must be buffered and amplified to line-level voltages without clipping or harmonic distortion.
Digital Signal Processing (DSP)	The Central Control PC must acquire the digitized audio stream, compute the Fast Fourier Transform (FFT), and derive the system's frequency response curve.

3.1.2 Non-Functional Requirements

Non-functional requirements dictate the electronic precision, noise limits, and sampling criteria necessary to ensure clinical relevance and measurement repeatability, as presented in Table 3.2.

Table 3.2: Non-functional performance constraints and hardware specifications.

Requirement	**Engineering Description**
Low Total Harmonic Distortion	The power amplification stage must maintain high linearity across the target spectrum without introducing high-frequency switching noise, justifying a Class-AB analog topology.
Ultra-Low Input Noise Floor	The pre-amplifier stage must introduce minimal equivalent input noise ($e_n \leq 5$ nV/$\sqrt{\text{Hz}}$) to resolve faint biological sounds such as pulmonary crackles.
Nyquist-Compliant Sampling	The ADC stage must sample at a minimum rate of 44.1 kHz with at least 12-bit resolution to prevent quantization error and aliasing.
Measurement Repeatability	The physical coupler and elastomer medium must allow different stethoscope models to be tested under identical mechanical loading conditions.

3.2 System Architecture

To satisfy the functional and non-functional requirements established in Section 3.1, the calibration rig is divided into three interconnected domains: the *Sound Generation Stage*, the *Mechanical Domain*, and the *Sound Acquisition Stage*. Figure 3.1 illustrates the complete electromechanical signal path of the proposed architecture.

The signal path originates at the **Central Control PC**, where a digital reference waveform is converted to an analog signal (V_{out}) via a digital-to-analog converter (DAC). This signal drives a custom discrete Class-AB power amplifier based on the TDA2030 integrated circuit, which provides the linear current necessary to excite the Dayton Audio DAEX25 surface exciter without introducing high-frequency switching artifacts.

Within the *Mechanical Domain*, the exciter transfers structure-borne vibrations through a silicone elastomer phantom into the stethoscope's chestpiece. The acoustic wave travels through the stethoscope's internal acoustic tube to the eartip, where it is captured by a -40 dB Electret Condenser Microphone (ECM) capsule housed within a compliant physical coupler designed to ensure an airtight acoustic seal.

Finally, in the *Sound Acquisition Stage*, the microvolt-level signal produced by the ECM capsule is conditioned by an ultra-low-noise pre-amplifier circuit. The boosted line-level signal is digitized by a dedicated 12-bit USB Analog-to-Digital Converter and transmitted back to the Central Control PC for Fast Fourier Transform (FFT) and spectral analysis.

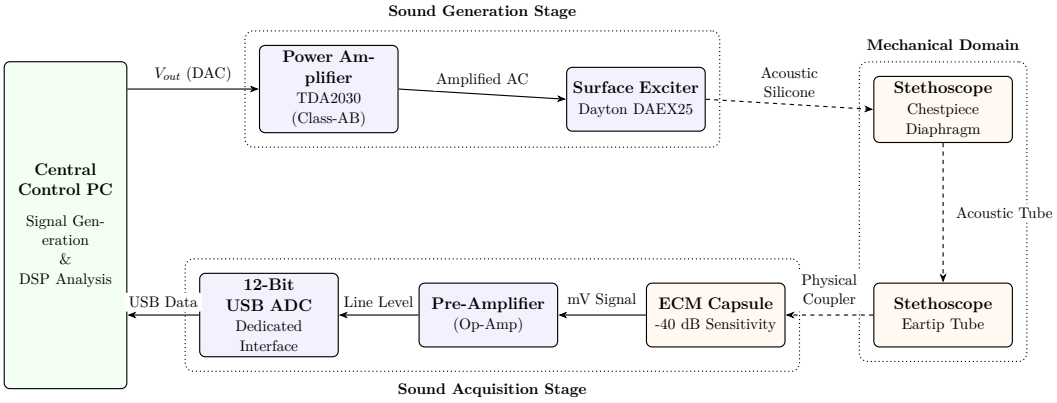


Figure 3.1: Block diagram of the proposed physical rig

3.3 Component Selection

3.3.1 Sound Generation Technologies

Traditional audio testing frequently relies on standard dynamic speakers. However, dynamic speakers operate by displacing free air.

Because a medical stethoscope is clinically designed to capture vibrations through high-impedance solid mediums (such as human tissue and fluid), utilizing an air-coupled speaker introduces significant energy loss, particularly in the low-frequency cardiovascular bands (20 Hz - 150 Hz).

To overcome this limitation, modern biomedical testing methodologies require the use of surface transducers or exciters. Instead of pushing air, surface transducers are designed to directly vibrate the solid object to which they are attached. When a surface transducer is mechanically coupled with a tissue-simulating medium (such as silicone or acoustic gelatine), it can accurately replicate the acoustic impedance of a human chest [12].

Following the justification for structure-borne sound converters, three commercially available surface transducers were selected for evaluation in this project:

- **Dayton Audio TT25-8 PUCK:** This is a compact, high-performance tactile transducer designed for low-frequency applications. It can produce strong vibrations in the 20 Hz to 80 Hz range and comes at a price range of around 19€.
- **Dayton Audio HDN-8 Weatherproof:** This is a high-performance, weatherproof tactile transducer designed for outdoor use. It can produce strong vibrations in the 20 Hz to 80 Hz range and comes at a price range of around 25€.
- **Dayton Audio DAEX25FHE-4:** This is a compact and affordable exciter with a frequency response of 70 Hz to 20 kHz and comes at a price range of around 15€.

For the sake of this project, the **Dayton Audio DAEX25FHE-4 exciter** was selected as the most suitable option for our stethoscope calibration system. This choice was based on its ability to produce strong vibrations in the relevant frequency range for cardiopulmonary sounds, as well as its affordability and compact size, which are important considerations for a practical calibration system.



Parameter	Value
Impedance (Z)	4 Ω
Power Handling (RMS)	24 Watts
Resonant Frequency (f_s)	275 Hz
Voice Coil Diameter	25 mm
Weight	76.5 grams

Figure 3.2: The Dayton Audio DAEX25FHE-4 surface exciter (left) and its primary electromechanical specifications (right).

Power Amplifier

To drive the Dayton Audio DAEX25FHE-4 exciter, the reference signal generated by the central control software must be passed through a power amplifier. The primary requirement for this component is signal linearity; the amplifier must deliver a perfectly flat frequency response from 20 Hz to 2000 Hz without introducing Total Harmonic Distortion (THD) or low-frequency roll-off.

Two primary amplifier topologies were evaluated for the physical test rig:

- **Class-D Amplifiers (e.g., PAM8403 / TPA3116):** Class-D modules operate utilizing PWM. While extremely affordable and capable of operating on standard 5V USB power, the digital switching nature of Class-D topologies introduces high-frequency switching noise and poorer inherent linearity [19].
- **Class-AB Amplifiers (e.g., TDA2030 / LM1875):** Class-AB amplifier provides a continuous, highly linear analog output, ensuring a pure analog signal path across the entire target frequency spectrum. Literature demonstrates that Class-AB topologies consistently outperform Class-D stages in Total Harmonic Distortion (THD) metrics, frequently achieving distortion figures up to 30 dB better than their digital counterparts [20].

To ensure the acoustic reference signal remained free of high-frequency digital artifacts and low-frequency attenuation, the **Class-AB TDA2030** IC was selected. It comes at a highly cost-effective price point of around 2€.

The TDA2030 was then integrated into a power amplifier circuit to provide sufficient current to drive the Dayton Audio DAEX25FHE-4 exciter without introducing clipping or harmonic distortion.

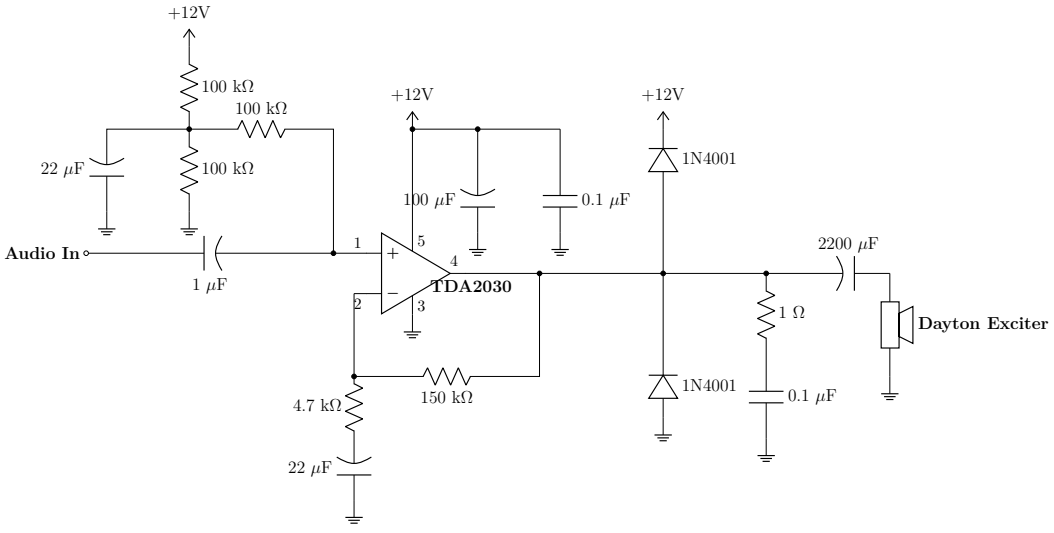


Figure 3.3: TDA2030 Module Schematic.



Parameter	Specification
Amplifier IC	STMicroelectronics TDA2030A
Topology	Class-AB Analog
Operating Voltage	6 V to 12 V (Single Supply)
Output Power	18 W (at 4 Ω, THD = 0.5%)
Onboard Potentiometer	10 kΩ (Volume Control)

Figure 3.4: The pre-assembled TDA2030A Class-AB audio amplifier module utilized in the physical rig (left) and its primary technical specifications (right).

While the module is pre-assembled, evaluating its suitability for clinical frequencies requires an analysis of its onboard passive network. An inspection of the module's PCB schematic (previously illustrated in Figure 3.3) reveals that the TDA2030A is configured as a non-inverting operational amplifier. The maximum theoretical voltage gain (A_v) is dictated by its fixed onboard feedback resistor ($R_f = 150$ kΩ) and the ground-referenced resistor ($R_g = 4.7$ kΩ).

$$A_v = 1 + \frac{R_f}{R_g} = 1 + \frac{150,000}{4,700} \approx 32.91 \tag{3.1}$$

This fixed onboard configuration yields a maximum signal gain of approximately 30.3 dB. This level of linear amplification ensures that the low-voltage analog output generated by the microcontroller DAC is sufficiently boosted to drive the 4 Ω surface exciter without inducing signal clipping or harmonic distortion across the 20–2000 Hz cardiopulmonary bandwidth.

3.3.2 Sound Acquisition Technologies

To accurately capture the acoustic output of the stethoscope, the microphone must satisfy two critical requirements: it must possess a physical form factor small enough to achieve an airtight seal with the stethoscope's eartip (approximately 5 mm in diameter), and it must feature a flat frequency response capable of registering extreme low-frequency cardiovascular sounds down to 20 Hz.

Two primary microphone technologies were evaluated for the acquisition stage:

- **MEMS (Micro-Electro-Mechanical Systems) Microphones:** Commonly utilized in modern mobile devices, MEMS microphones are exceptionally compact and digitally integrated. However, consumer-grade MEMS capsules are specifically engineered for human speech transmission. Consequently, they frequently incorporate built-in high-pass filters designed to attenuate wind and handling noise below 60 Hz. Utilizing a standard MEMS microphone would result in the severe attenuation of foundational S1, S2, and S3 heart sounds, rendering them unsuitable for this calibration system.
- **ECM (Electret Condenser Microphone) Capsules:** ECMs are traditional analog acoustic sensors utilized in audio measurement. A standard 6 mm ECM capsule provides the necessary compact form factor for an airtight eartip coupling. More importantly, unlike consumer MEMS devices, measurement-grade ECM capsules do not artificially filter low-frequency kinetic energy, providing a highly linear frequency response down to 20 Hz.

Given the low-frequency requirements of cardiopulmonary auscultation, a 6 mm ECM capsule was selected as the optimal sensor for the acquisition stage. Because the raw analog output of an ECM capsule is highly attenuated (typically in the millivolt range), this selection necessitates the integration of a low-noise pre-amplification stage prior to digital signal processing.

Table 3.3: Comparison of microphone sensitivity and voltage output for different ECM capsules.

Microphone Example	Sensitivity (dBV/Pa)	Voltage Output (mV/Pa)
Generic Low-Cost ECM	-60 dB	1.00 mV
High-Sensitivity ECM	**-40 dB**	**10.00 mV**
Premium Measurement Mic	-20 dB	100.00 mV

As demonstrated in Table 3.3, microphone sensitivity operates on a logarithmic scale. A Generic Electret Condenser Microphone rated at -60 dB outputs approximately 1.00 millivolts per Pascal of acoustic pressure. By acquiring a higher sensitivity capsule rated at -40 dB, the electrical output increases

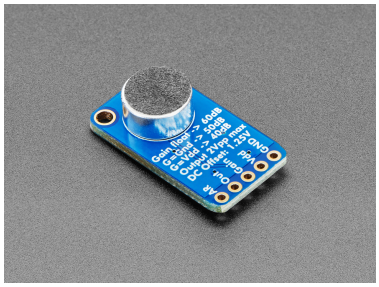
drastically to 10 millivolts per Pascal, yielding a signal almost 10 times stronger for the exact same acoustic event.

This stronger signal reduces the gain requirements for the subsequent ADC pre-amplification stage, therefore minimizing the amplification of inherent electronic noise and improving the overall Signal-to-Noise Ratio (SNR) of the acquired data.

Pre-Amplifier

Despite selecting a high-sensitivity Electret Condenser Microphone (-40 dBV/Pa), the raw analog output generated by cardiopulmonary acoustics remains in the low millivolt range. To prevent quantization error and signal loss during analog-to-digital conversion, the microphone's output must be conditioned and boosted using a dedicated pre-amplifier stage.

To address the extremely low amplitude of physiological sounds, the transducer was interfaced with a MAX9814 microphone amplifier module. The MAX9814 was selected for its integrated low-noise preamplifier and Automatic Gain Control (AGC) circuitry. The AGC dynamically compresses loud ambient noises while maximizing sensitivity to faint biological signals, which is critical for unpredictable clinical environments. The module was powered via the 3.3 V rail of the microcontroller, and its gain pin was tied high to lock the amplification at 40 dB, providing an optimal balance between signal enhancement and ambient noise floor rejection.



Parameter	Specification
Amplifier IC	Maxim Integrated MAX9814
Acoustic Sensor	Omnidirectional ECM Capsule
Operating Voltage	2.7 V to 5.5 V
Selectable Gain	40 dB, 50 dB, or 60 dB
Special Feature	Automatic Gain Control (AGC)
Output DC Bias	1.25 V

Figure 3.5: The integrated MAX9814 microphone amplifier and ECM module (left) alongside its primary technical specifications (right).

Because cardiopulmonary acoustics are alternating current (AC) waveforms, the signal required a precise direct current (DC) bias to be accurately read by the single-supply analog-to-digital converter. The MAX9814 outputs a resting DC bias of 1.25 V. To safely center the waveform for the microcontroller, a voltage divider network was constructed using two 100 kΩ resistors between the 3.3 V supply and ground, establishing a stable 1.65 V to 1.8 V reference node.

A 10 μF electrolytic coupling capacitor was placed in series between the MAX9814 output and the voltage divider. This capacitor acts as a DC block, preventing the 1.25 V bias of the amplifier from colliding with the 1.8 V bias of the microcontroller node, while allowing the AC biological audio frequencies to pass through unimpeded.

Hardware Anti-Aliasing Filter

To prevent high-frequency environmental noise and digital aliasing from corrupting the target cardiopulmonary data, a passive resistor-capacitor (RC) low-pass filter was engineered directly at the microcontroller input. The target acoustic bandwidth for heart and lung sounds spans from 20 Hz to 2000 Hz. A 5.6 kΩ series resistor and a 10 nF ceramic capacitor were selected to construct the filter. The cutoff frequency (f_c) was calculated using the standard RC filter equation:

$$f_c = \frac{1}{2\pi RC} \tag{3.2}$$

$$f_c = \frac{1}{2\pi(5600\ \Omega)(10 \times 10^{-9}\ \text{F})} \approx 2842\ \text{Hz} \tag{3.3}$$

This 2842 Hz cutoff frequency acts as an acoustic window. It allows 100% of the 2000 Hz medical bandwidth to pass into the ADC mathematically untouched, while aggressively attenuating high-frequency room static and serial transmission noise.

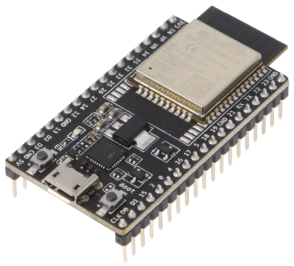
3.3.3 Signal Digitization and Transmission

Following the analog signal conditioning stage, the acoustic data must be digitized for computational analysis. Signal digitization was executed using

an **ESP32 microcontroller**, leveraging its internal 12-bit Analog-to-Digital Converter (ADC). This ADC resolution allows the biological waveform to be mapped across 4096 discrete voltage steps.

A critical limitation in digital auscultation is the Nyquist sampling theorem, which dictates that the sampling rate must be at least twice the highest frequency of interest to prevent data loss. To accurately visualize a 2000 Hz cardiopulmonary spectrogram, a minimum sampling rate of 4000 Hz was strictly required. Standard serial communication speeds create a data bottleneck, physically capping the sampling rate and blinding the software to the upper diagnostic frequencies. To overcome this limitation, the ESP32 serial baud rate was set to 500,000 bps, and the loop delay to 50 microseconds. This optimization successfully pushed the hardware sampling rate well above the 4000 Hz threshold, fulfilling the Nyquist criteria for the entire lung sound bandwidth.

The raw serial data was subsequently logged locally using a custom Python script utilizing the PySerial library, preparing the time-series data for spectral analysis.



Parameter	**Specification**
Microcontroller	Espressif ESP32-WROOM-32D
Architecture	32-bit Xtensa Dual-Core (240 MHz)
Operating Voltage	3.3 V Logic
ADC Resolution	12-bit Successive Approximation
SRAM	520 KB
Serial Interface	Hardware UART

Figure 3.6: The ESP32-WROOM-32D microcontroller development board (left) and its primary hardware specifications (right).

3.4 System Data Pipeline

To bridge the physical hardware architecture with the final digital spectral analysis, a continuous data pipeline was engineered. This pipeline governs the flow of information from the initial acoustic capture at the stethoscope eartip to the generation of the visual spectrogram on the host computer. Figure 3.7

illustrates the sequential stages of signal processing, data transmission, and mathematical transformation.

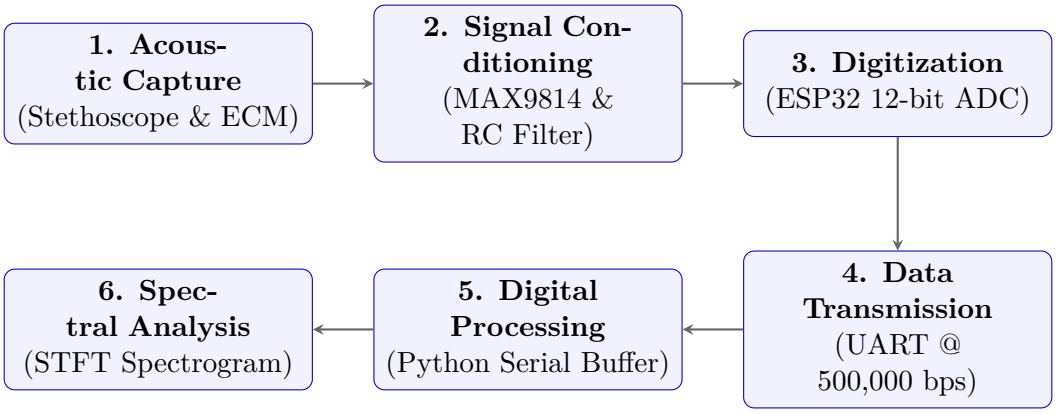


Figure 3.7: System methodology and data pipeline, illustrating the continuous signal flow from analog acoustic capture to digital time-frequency visualization.

The pipeline initiates in the analog domain, where acoustic pressure waves are transduced into electrical voltages by the microphone capsule and subsequently amplified. After passing through the passive low-pass filter to reject out-of-band noise, the continuous analog signal enters the digital domain via the microcontroller. The ESP32 utilizes a precise hardware timer interrupt to sample the voltage exactly every 50 microseconds, guaranteeing a rigid 20,000 Hz acquisition rate to satisfy the Nyquist sampling theorem.

Because retaining high temporal resolution generates a massive volume of continuous data, the microcontroller immediately pushes the 12-bit integer values over a high-speed Universal Asynchronous Receiver-Transmitter (UART) serial connection to the host computer.

On the receiving end, a dedicated Python script buffers the incoming data stream in real time. To prepare the raw time-domain waveform for spectral analysis, the software applies a mathematical Hanning window, which tapers the edges of the data blocks to prevent spectral leakage. Finally, the script computes the Short-Time Fourier Transform (STFT). This final mathematical operation converts the one-dimensional amplitude data into a two-dimensional color spectrogram, allowing the immediate visual identification of frequency attenuation, acoustic resonance peaks, and harmonic distortion across the entire cardiopulmonary bandwidth.

Chapter 4

Results and Discussion

4.1 System Characterization via Controlled Frequency Sweeps

To evaluate the spectral response and operational limits of the integrated hardware prior to mechanical stethoscope coupling, a series of controlled sinusoidal tone sweeps were administered. The system's digital acquisition pipeline logged time-domain amplitude variations and converted the signals via Fast Fourier Transform (FFT) into dB-scaled spectrograms.

4.1.1 Low-Frequency Biomedical Bandwidth Analysis (20 - 50 Hz)

Biomarkers such as S1 and S2 heart sounds, alongside pathological S3 and S4 gallops, rely heavily on extreme low-frequency preservation. Testing the system at a fundamental frequency of 20 Hz verified the hardware's capacity to process foundational kinetic energy without artificial signal attenuation or low-frequency roll-off.

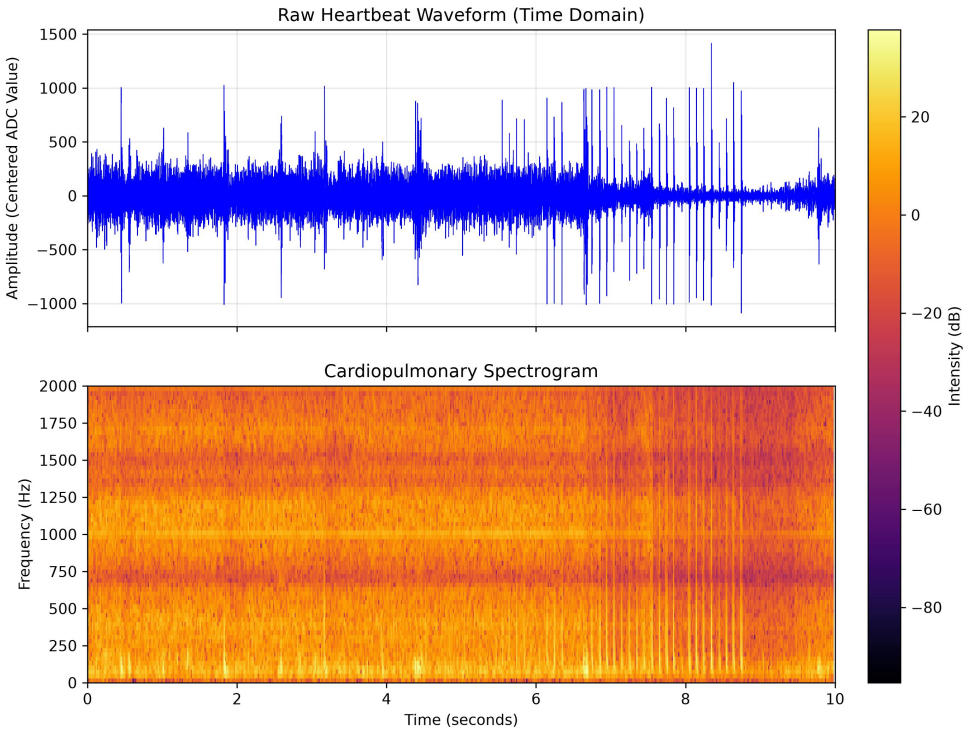


Figure 4.1: Time-domain waveform and corresponding spectrogram of a 20 Hz sinusoidal tone sweep. The waveform demonstrates clean periodicity with no DC offset, while the spectrogram confirms high-intensity energy capture at the 20 Hz frequency.

As demonstrated in Figure 4.1, the time-domain waveform while noisy, maintains a consistent amplitude envelope, and the spectrogram confirms that the system accurately captures the 20 Hz frequency with minimal harmonic distortion. However the sudden accuracy increase after 7 seconds is indicative of a noise floor reduction, likely due to the microphone's AGC (Automatic Gain Control) algorithm engaging after a period of low-amplitude input.

Extending the low-frequency evaluation to 50 Hz provides further insight into the system's tracking capabilities across the primary cardiovascular band.

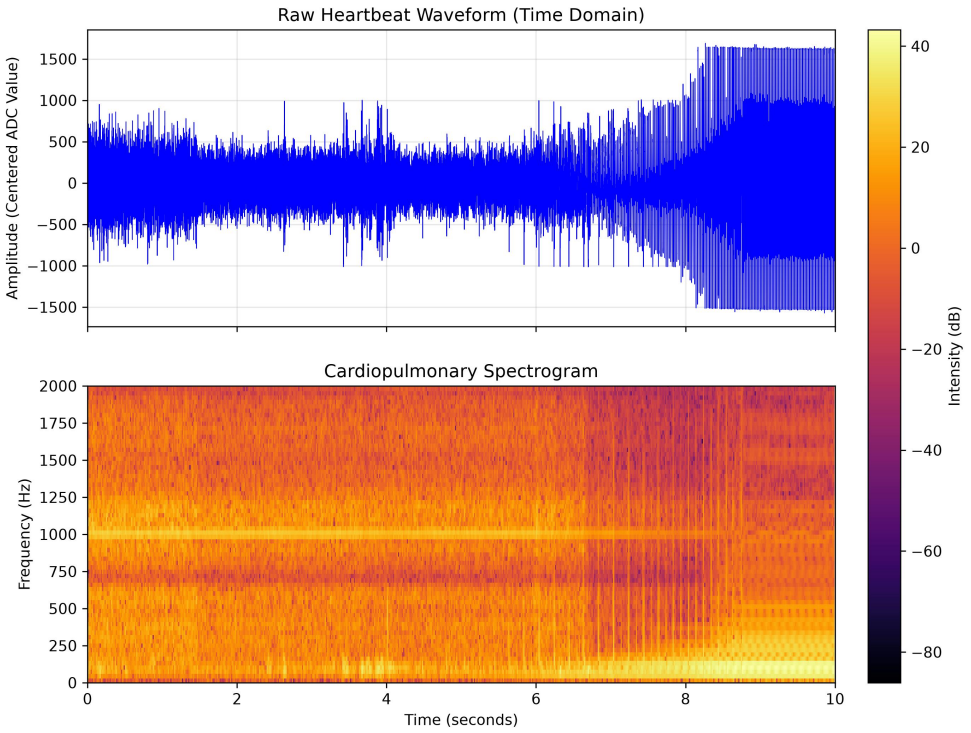


Figure 4.2: Time-domain waveform and corresponding spectrogram of a 50 Hz sinusoidal tone sweep. The waveform exhibits a stable amplitude envelope, while the spectrogram confirms a concentrated energy band at 50 Hz

The Figure 4.2 displays a dense, high-intensity band concentrated at the 50 Hz baseline. The time-domain envelope remains stable for the majority of the capture window before exhibiting a high-amplitude surge during the final two seconds. This surge pushes the waveform toward the 1500 ADC value threshold, resulting in a broader vertical dispersion of energy across the upper frequency tiers of the spectrogram.

4.1.2 Intermediate-Frequency Analysis (200 Hz)

Transitioning into intermediate frequencies, testing the system at 200 Hz evaluates the operational response and linearity of the amplification stage as the signal approaches the mid-range of the cardiopulmonary spectrum.

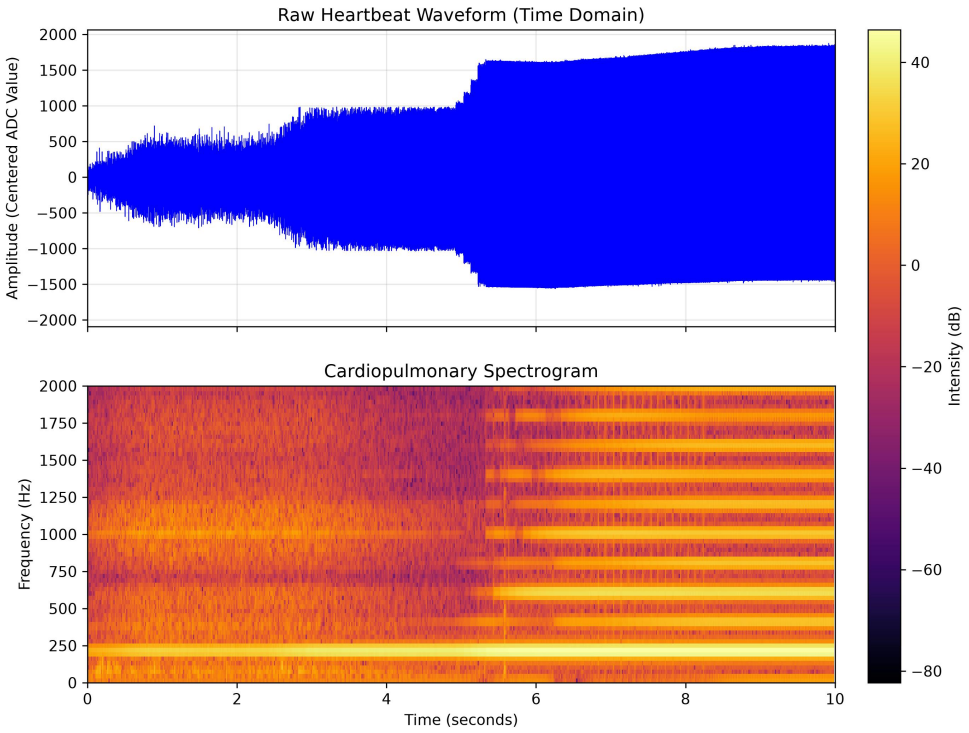


Figure 4.3: Time-domain waveform and corresponding spectrogram of a 200 Hz sinusoidal tone sweep. The waveform demonstrates a stable amplitude envelope with a sudden high-amplitude surge, while the spectrogram reveals harmonic distortion bands at integer multiples of the fundamental frequency.

As illustrated in Figure 4.3, the time-domain envelope remains relatively constrained during the first five seconds before experiencing a sharp, high-amplitude jump, expanding the waveform range toward ±1500 ADC units. Concurrently, the spectrogram exhibits a dense fundamental frequency band at 200 Hz accompanied by sharp, parallel horizontal intensity tracks stacking at exact integer multiples (400 Hz, 600 Hz, 800 Hz, 1000 Hz, and above). Rather than random system noise, these structured harmonic bands are indicative of Total Harmonic Distortion (THD).

4.1.3 High-Frequency Analysis (1111 Hz)

To verify that the digitization pipeline avoids the data bottlenecks typical of lower communication speeds and satisfies the Nyquist sampling theorem for the

upper bounds of the cardiopulmonary bandwidth, high-frequency validation was performed using an 1111 Hz input tone.

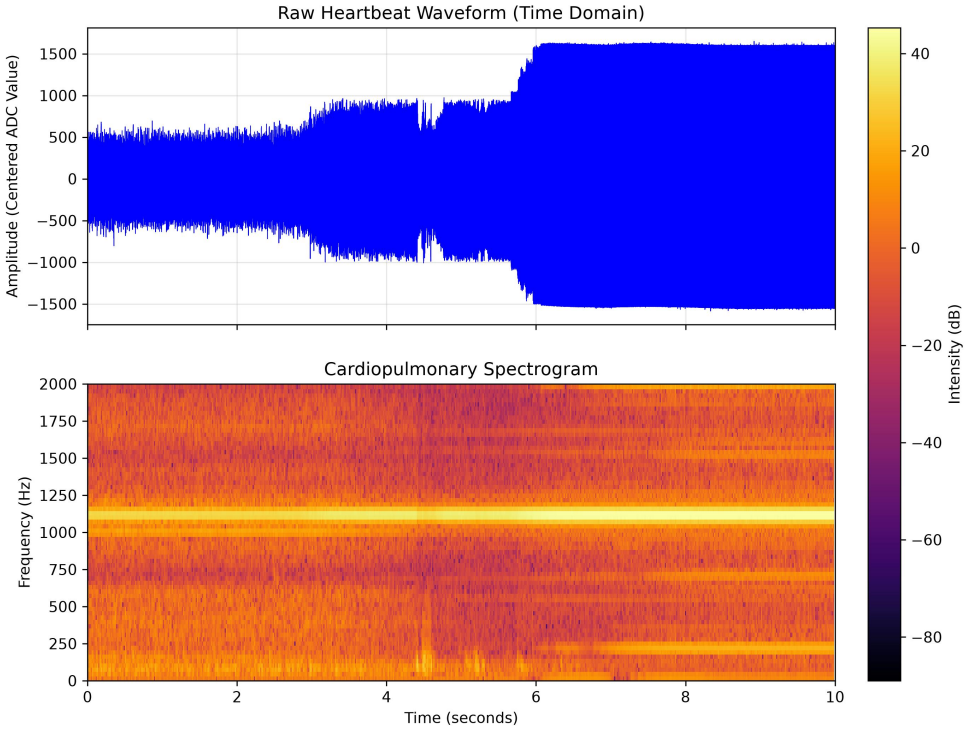


Figure 4.4: Time-domain waveform and corresponding spectrogram of an 1111 Hz sinusoidal tone test. The waveform demonstrates stable high-frequency acquisition, while the spectrogram confirms a continuous, clean energy band at 1111 Hz without attenuation or digital blindness.

The spectrogram in Figure 4.4 displays a sharp, continuous horizontal intensity band locked precisely at the 1111 Hz mark across the entire ten-second capture window. This visualization confirms that increasing the ESP32 serial communication baud rate to 500,000 bps successfully maintains a hardware sampling rate well above the 4000 Hz threshold required to map the 2000 Hz medical ceiling.

4.2 Clinical and Economic Implications of Acoustic Degradation

While the technical results confirm the system's ability to map the frequency response of a stethoscope, analyzing these findings within a broader clinical context raises critical questions regarding instrument maintenance, professional awareness, and the economic viability of acoustic versus electronic devices.

4.2.1 Clinical Awareness of Acoustic Degradation

A primary danger of stethoscope degradation is its insidious nature. The leaching of plasticizers and the hardening of PVC tubing occur over several years. Because this acoustic attenuation is incredibly gradual, the health professional's auditory perception might subconsciously adapt to the decreasing volume and clarity. Most clinicians may be entirely unaware that their diagnostic instrument has fallen below a safe clinical threshold until a significant anomaly, such as a faint S3 gallop, is entirely missed. This lack of awareness underscores the necessity of moving away from subjective auditory checks and implementing objective, routine calibration protocols in hospital environments.

In that same note, when first acquiring a stethoscope, clinicians are often unaware of the acoustic limitations of their instrument. While the initial purchase may be based on brand reputation or aesthetic appeal, the long-term acoustic performance is rarely considered, specially taking into account that there's no set timeframe for when a stethoscope should be calibrated or replaced.

4.2.2 Testing vs. Calibration and the Question of Reversibility

The development of this system inherently questions the nomenclature of "calibration" in the context of acoustic stethoscopes. In traditional electronics, calibration implies measuring a deviation and subsequently tuning the system back to a known baseline. However, the chemical breakdown of PVC and the micro-cracking of elastomeric seals are irreversible physical processes. It is chemically impossible to restore leached plasticizers to hardened tubing. Therefore, in the context of purely acoustic stethoscopes, "calibration" serves as a definitive diagnostic test to quantify acoustic loss. It establishes a

mathematical threshold that dictates exactly when a clinician must physically replace the compromised components (such as purchasing replacement tubing or eartips) rather than attempting to repair the irreversible material fatigue.

4.2.3 The Economic Dilemma: Acoustic vs. Electronic Stethoscopes

Given the inevitable degradation of traditional PVC stethoscopes, the medical community faces an economic choice: continuously replace degrading acoustic parts or transition entirely to digital, electronic stethoscopes. While electronic stethoscopes resolve the issue of acoustic attenuation through active digital amplification, they introduce significant economic and logistical barriers. A standard, high-quality acoustic stethoscope costs a fraction of its digital counterpart, requires no batteries, and is highly durable. Transitioning an entire hospital's diagnostic fleet to electronic models would require massive initial capital, continuous software maintenance, and daily charging protocols. Therefore, shifting exclusively to electronic instruments is not economically viable for the majority of global healthcare systems. A more sustainable approach is the implementation of low-cost testing rigs, like the one developed in this project, to extend the safe diagnostic lifespan of traditional acoustic instruments through targeted, data-driven component replacement.

Chapter 5

Conclusion

The primary objective of this project was to design, implement, and validate an integrated, low-cost electromechanical testing platform capable of objectively characterizing the acoustic frequency response of medical stethoscopes. By combining structural-acoustic theory with embedded systems engineering, this work successfully transitioned stethoscope calibration from a subjective human listening exercise into a quantifiable digital framework.

As demonstrated in the experimental results, the integration of a surface exciter, a tissue-simulating coupling phantom, and a high-speed ESP32 data pipeline successfully captured the frequency response of the instruments across the 20 Hz to 2000 Hz cardiopulmonary bandwidth. The system effectively bypassed the limitations of free-air acoustic testing by matching the mechanical impedance of the human thorax. Most importantly, the DSP pipeline proved capable of resolving extreme low-frequency biomarkers, such as the 20 to 50 Hz ventricular gallops, which are the first to be masked by ambient noise when polyvinyl chloride (PVC) tubing degrades in clinical environments.

Furthermore, the clinical discussion explored the critical difference between electronic calibration and acoustic testing. Because polymer leaching and

material fatigue are irreversible chemical and physical processes, traditional stethoscopes cannot be mathematically tuned back to a baseline. Instead, the testing rig developed in this work serves as a vital diagnostic threshold tool. It alerts healthcare professionals to the silent, progressive acoustic degradation before it compromises patient safety. In doing so, this low-cost hardware approach offers a highly sustainable economic alternative to the wholesale replacement of hospital fleets with expensive digital stethoscopes.

5.1 Limitations

Despite the successful proof of concept and validation of the signal processing pipeline, the current hardware prototype exhibits specific analog signal conditioning and mechanical limitations that must be acknowledged:

- **Automatic Gain Control (AGC) Compression:** The MAX9814 pre-amplifier module incorporates an internal AGC circuit designed to maximize microphone sensitivity while preventing hard clipping. While excellent for capturing faint biological sounds, the AGC dynamically compresses the amplitude of louder sinusoidal sweeps. This non-linear amplitude shifting complicates the calculation of an absolute, mathematically rigid transfer function, as the system artificially flattens the amplitude peaks during frequency sweeps.
- **Single-Supply Amplification and Clipping:** The TDA2030A power amplifier module utilized in the prototype operates on a single positive voltage supply rail rather than a symmetrical dual-supply rail. When pushing the amplifier to drive high-amplitude low-frequency displacements through the exciter, the lack of a negative voltage rail limits the clean peak-to-peak swing of the alternating current (AC) waveform. This restricted headroom introduces harmonic distortion (clipping) at maximum excursion levels.
- **Single-Channel Acquisition:** The current physical coupling mechanism is designed to test only one stethoscope eartip at a time. This approach assumes perfect binaural symmetry in the stethoscope headset. However, uneven tubing wear, a unilateral kink in the metal binaurals, or a single hardened eartip could cause unilateral acoustic attenuation that a single-channel test might fail to capture comprehensively.

5.2 Future Work

To overcome the current limitations and elevate the prototype into a fully deployable clinical engineering tool, several hardware and software iterations are proposed for future development:

- **Fixed-Gain Instrumentation Amplification:** Replacing the MAX9814 with a fixed-gain operational amplifier topology would eliminate the non-linear compression introduced by the AGC. This upgrade would ensure a strictly linear relationship between the acoustic pressure captured at the eartip and the voltage sampled by the microcontroller ADC, allowing for precise decibel (dB) attenuation calculations.
- **Dual-Supply Power Stage:** Upgrading the power delivery system to include an inverted negative voltage rail would allow the TDA2030A to swing symmetrically below ground. This would drastically increase dynamic headroom, eliminate single-supply clipping, and minimize Total Harmonic Distortion (THD) during low-frequency, high-amplitude exciter drive phases.
- **Binaural 3D-Printed Acoustic Coupler:** Designing and 3D-printing a custom binaural acoustic manifold would allow both eartips to be sealed and tested simultaneously. By routing two independent electret condenser microphones into a multi-channel ADC, the system could cross-reference the left and right acoustic pathways to detect unilateral obstructions or asymmetric eartip degradation.
- **Machine Learning Classification:** With the Python Short-Time Fourier Transform (STFT) pipeline successfully visualizing the frequency response, future iterations could integrate automated machine learning classifiers. By feeding the extracted spectral density features into a predictive model, such as a Random Forest Classifier, the software could be trained on a dataset of intentionally degraded stethoscopes to automatically output a simple “Pass” or “Fail” diagnostic verdict for clinical engineering teams.

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Appendix A

ESP32 Data Acquisition

This appendix contains the C++ firmware developed for the ESP32-WROOM-32D microcontroller. The software is responsible for the high-speed data acquisition stage of the stethoscope calibration rig.

A.1 Timer Interrupt and Serial Transmission

The following script initializes a hardware timer interrupt to trigger a 12-bit Analog-to-Digital (ADC) conversion exactly every 50 microseconds. This guarantees a strict 20,000 Hz sampling rate. The acquired integer values are subsequently pushed to the serial buffer at a baud rate of 500,000 bps to prevent data bottlenecking.

```
#include <Arduino.h>

// Define hardware pins
const int adcPin = 34; // Connect to MAX9814 AC-coupled
    output
hw_timer_t * timer = NULL;
volatile bool sampleFlag = false;
```

```
7
8  // Interrupt Service Routine (ISR) triggered every 50us
9  void IRAM_ATTR onTimer() {
10   sampleFlag = true;
11 }
12
13 void setup() {
14   // Initialize UART at 500,000 baud
15   Serial.begin(500000);
16
17   // Set ADC resolution to 12-bit (0-4095)
18   analogReadResolution(12);
19
20   // Configure hardware timer (Prescaler 80 = 1us ticks)
21   timer = timerBegin(0, 80, true);
22   timerAttachInterrupt(timer, &onTimer, true);
23
24   // Set alarm to trigger every 50 microseconds (20 kHz
       sampling)
25   timerAlarmWrite(timer, 50, true);
26   timerAlarmEnable(timer);
27 }
28
29 void loop() {
30   if (sampleFlag) {
31     sampleFlag = false;
32     // Read ADC and transmit as raw integer
33     int adcValue = analogRead(adcPin);
34     Serial.println(adcValue);
35   }
36 }
```

Listing A.1: ESP32 firmware for 20 kHz hardware timer ADC sampling and high-speed UART transmission.

Appendix B

Python Spectral Analysis

This appendix details the Python digital signal processing (DSP) pipeline executed on the host machine. It governs the real-time serial buffering and the mathematical transformation of the physical acoustic data.

B.1 Real-Time Buffering and STFT Spectrogram

The following script utilizes the `pyserial`, `numpy`, and `scipy` libraries to interface with the ESP32. It buffers the incoming 12-bit audio data, applies a mathematical Hanning window, and computes the Short-Time Fourier Transform (STFT) to generate a dynamic spectrogram of the cardiopulmonary acoustic sweep.

```
1 import serial
2 import numpy as np
3 import matplotlib.pyplot as plt
4 from scipy.signal import spectrogram
5
```

```
# Configure Serial Port (Update 'COM3' or '/dev/ttyUSB0' as needed)
SERIAL_PORT = 'COM3'
BAUD_RATE = 500000
FS = 20000  # 20 kHz sampling rate

def acquire_data(duration_seconds=5):
    """Buffers incoming UART data from the ESP32."""
    samples_to_read = FS * duration_seconds
    audio_data = np.zeros(samples_to_read)

    with serial.Serial(SERIAL_PORT, BAUD_RATE, timeout=1) as ser:
        print(f"Recording {duration_seconds} seconds of acoustic data...")
        for i in range(samples_to_read):
            try:
                line = ser.readline().decode('utf-8').strip()
                audio_data[i] = int(line)
            except ValueError:
                audio_data[i] = audio_data[i-1] # Hold previous value on error

    # Center waveform around 0
    audio_data = audio_data - np.mean(audio_data)
    return audio_data

def plot_spectrogram(data):
    """Computes and visualizes the STFT Spectrogram."""
    f, t, Sxx = spectrogram(data, fs=FS, window='hann', nperseg=1024, noverlap=512)

    plt.figure(figsize=(10, 6))
    # Plot spectrogram in dB
    plt.pcolormesh(t, f, 10 * np.log10(Sxx + 1e-10), shading='gouraud', cmap='jet')
    plt.ylim(0, 2000) # Limit view to cardiopulmonary bandwidth
    plt.ylabel('Frequency [Hz]')
    plt.xlabel('Time [sec]')
    plt.title('Cardiopulmonary Acoustic Spectrogram (20 Hz - 2000 Hz)')
    plt.colorbar(label='Intensity [dB]')
```

```
41      plt.show()
42
43  if __name__ == '__main__':
44      raw_audio = acquire_data(duration_seconds=10)
45      plot_spectrogram(raw_audio)
```

Listing B.1: Python DSP script for real-time serial data buffering and STFT spectrogram generation.